

LEC31 - Ratio and Root Tests

Warm Up Problem

Suppose $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{5}$. What can we say about

the convergence of $\sum_{n=1}^{\infty} a_n$

$\Rightarrow$ Converges absolutely

The Ratio Test

Consider the series $\sum_{n=1}^{\infty} a_n$ where a_n is never 0. Let $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = p$

(if it exists or is ∞)

i) $0 \leq p < 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ converges absolutely

ii) $p > 1$ or $p = \infty \Rightarrow \sum_{n=1}^{\infty} a_n$ diverges

iii) $p = 1 \Rightarrow$ Inconclusive

Example

Is $\sum_{n=1}^{\infty} \frac{n \cdot (2n)!}{3^n (n-1)!}$ absolutely convergent, conditionally convergent, or divergent?

$\Rightarrow$ We proceed using the ratio test

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{(n+1)(2n+2)!}{3^{n+1}(n+1)!} \cdot \frac{3^n (n-1)!}{n(2n)!} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)(2n+2)(2n+1)(2n)! \cdot 3^n \cdot (n-1)!}{3 \cdot 3^n \cdot (n+2)(n+1)! \cdot n \cdot (2n)!}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{(n+1)(2n+2)(2n+1)}{3n(n+2)} = \infty$$

$\Rightarrow$ Therefore, the series diverges

The Root Test

Consider the series $\sum_{n=1}^{\infty} a_n$, where a_n is never 0. Let $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = p$

(if it exists or is ∞)

i) $0 \leq p < 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ converges absolutely

ii) $p > 1$ or $p = \infty \Rightarrow \sum_{n=1}^{\infty} a_n$ diverges

iii) $p = 1 \Rightarrow$ inconclusive

Example

Suppose $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \frac{1}{3}$. What can we say about the convergence of $\sum_{n=1}^{\infty} 2^n a_n$?

$\Rightarrow$ Proceed using the root test

$$\Rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{|2^n a_n|} = \lim_{n \rightarrow \infty} 2 \sqrt[n]{|a_n|} = \frac{2}{3}$$

$\Rightarrow$ Therefore, the series converges

Choosing a Convergence Test

1. Is it a series we know? $\Rightarrow \frac{1}{n}, \frac{1}{n^2}$, etc.
2. Alternating $\Rightarrow$ Alternating series test
3. Similar to p-series or geometric $\Rightarrow$ (Limit) comparison test
4. Factorials or powers $\Rightarrow$ Ratio test
5. $\lim_{n \rightarrow \infty} a_n \neq 0 \Rightarrow$ Divergence Test
6. Last resort $\Rightarrow$ Check for telescoping, integral test

Example

Which test will help determine if the series $\sum_{n=1}^{\infty} \frac{n^4 + 5n^2}{2n^3 + 30}$ converges?

$\Rightarrow$ Similar to p-series $\sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2}$

$\Rightarrow$ Therefore, try limit comparison test

Example

Which test will help determine if the series $\sum_{n=1}^{\infty} \frac{2^n}{(n-1)!}$ converges?

$\Rightarrow$ Powers and factorials

$\Rightarrow$ Try ratio test

Example

Which test will help determine if the series $\sum_{n=1}^{\infty} \frac{1}{n \arctan(n)}$ converges?

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n \arctan(n)} \neq 0$$

$\Rightarrow$ Therefore, the series diverges by the divergence test

Example

Which test will help determine whether the series $\sum_{n=1}^{\infty} (-1)^n \frac{3^n}{5^n}$ converges?

$\Rightarrow$ We know $\sum_{n=1}^{\infty} \left(\frac{3}{5}\right)^n$ converges, which implies that $\sum_{n=1}^{\infty} (-1)^n \left(\frac{3}{5}\right)^n$

also converges

Example

Which test will help determine if the series $\sum_{n=1}^{\infty} \frac{(-0.8)^n}{n}$ converges?

$\Rightarrow$ Could use alternating series test for $(-1)^n \frac{(0.8)^n}{n}$, or compare to $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$, which converges